

Lecture 4: Beta-Representation of Returns and Factor Models

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Intro

- All that matters for pricing is $\text{Proj}(m|\mathcal{R}) \implies$ many SDFs may have the same pricing implications
- Hansen and Jagannathan ([1991](#)) explored one SDF that is affine in returns, but it has lots of parameters
- Is there a more parsimonious representation for the SDF?

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- Hansen and Jagannathan (1991) explored one SDF that is affine in returns, but it has lots of parameters
- Is there a more parsimonious representation for the SDF?
- In industry and MBA classes, people talk a lot about the CAPM
- How is that related to all this SDF stuff we are talking?
- What about Fama-French factors? Factor investing? Are these just buzzwords?

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Let $\beta_R \equiv \frac{\text{Cov}(m, R)}{\sigma^2(m)}$ and $\lambda \equiv -\frac{\sigma^2(m)}{\mathbb{E}[m]}$

$$\mathbb{E}[R - R_f] = \beta_R \cdot \lambda$$

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- Why do assets have different risk premia? Because they have different betas with respect to the SDF
- If you use the conditional pricing equation, a conditional version holds:

$$\mathbb{E}_t[R_{t+1} - R_{f,t}] = \beta_{R,t} \cdot \lambda_t$$

The CAPM as a Special Case

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Since m must price R_m and R_f , we have two equations in two unknowns (a, b) :

$$\begin{cases} \mathbb{E}[mR_f] = 1 \\ \mathbb{E}[mR_m] = 1 \end{cases} \implies \begin{cases} a = \frac{1}{R_f} + b\mathbb{E}[R_m] \\ b = \frac{\mathbb{E}[R_m] - R_f}{\sigma^2(R_m)R_f} \end{cases}$$

Exercise: verify this!

The CAPM As A Special Case

Plugging back in, we have

$$m = \frac{1}{R_f} \left(1 + \frac{\mathbb{E}[R_m] - R_f}{\sigma^2(R_m)} (\mathbb{E}[R_m] - R_m) \right)$$

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Calculating the beta, we have

$$\beta_R \equiv \frac{\text{Cov}(a - bR_m, R)}{\sigma^2(a - bR_m)} = -\frac{\text{Cov}(R_m, R)}{b \cdot \sigma^2(R_m)} \equiv -\frac{1}{b} \cdot \beta_R^{\text{Market}}$$

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and now the λ :

$$\lambda \equiv -\frac{\sigma^2(m)}{\mathbb{E}[m]} = b^2 \cdot \frac{\sigma^2(R_m)}{\mathbb{E}[m]} = -b \cdot \mathbb{E}[R_m - R_f]$$

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Then:

$$\mathbb{E}[R - R_f] = -\frac{1}{b} \cdot \beta_R \cdot (-b \cdot \mathbb{E}[R_m - R_f]) = \beta_R \cdot \mathbb{E}[R_m - R_f]$$

- Last slide: all we need to justify the CAPM is a way to justify $m = a - bR_m$
- Many ways to do this:
 - Quadratic utility in two periods
 - Exponential utility in two periods and log-normally distributed returns
 - Infinite horizon, quadratic utility, and i.i.d. returns...
- Most importantly: a theory about where returns come from implies a theory about the SDF
- The converse is obviously also true

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- This idea actually generalized to more sources of risk

Definition

Returns follow a *beta-representation* if there is $\gamma \in \mathbb{R}$ and $\lambda \in \mathbb{R}^K$ such that, for any return R_i , we can find $\beta_i \in \mathbb{R}^K$ ensuring that

$$\mathbb{E}[R_i] = \gamma + \beta_i' \lambda$$

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- Very important: only β_i depends on the asset i
- γ and λ are economy-wide parameters
- If the returns are excess returns, then $\gamma = 0$. Why?

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- Because they are very general: Roll ([1977](#))
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- Because they are very general: Roll (1977)
- If there exists a mean-variance frontier of returns, there is a beta-representation
- All returns are of the form $R_i = R^* + w_i R^{*e} + n_i$, with $\mathbb{E}[n_i] = 0$
 - R^* is related to the Hansen-Jagannathan SDF
 - $R^{*e} = \text{Proj}(1|\mathcal{R}^e)$, where $\mathcal{R}^e \equiv$ the space of excess returns
 - w_i is a scalar, and n_i is some excess return
 - All three elements are orthogonal
- Mean-variance efficient returns are given by $R_i = R^* + w_i R^{*e}$

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 - All three elements are orthogonal
- Mean-variance efficient returns are given by $R_i = R^* + w_i R^{*e}$
- Upside: we don't lose much thinking about beta-representation of returns
- Downside: using a beta representation imposes almost no structure on returns, no discipline

Beta-Representation and Factor Models

- Beta-representations are closely related to factor models of returns
- A K -factor model of returns is given by

$$R_{i,t} = \alpha_i + \beta_i' F_t + \epsilon_{i,t}$$

where $F_t = (f_{1,t}, \dots, f_{K,t})^\top$ and $\epsilon_{i,t}$ is uncorrelated with F_t .

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- If we take the expectation: $\mathbb{E}[R_i] = \alpha_i + \beta_i' \mathbb{E}[F]$
- A beta-representation would also require $\alpha_i = \alpha$ for all i ;
- Factor models are everywhere in Macro and Finance: common movements are driven by few factors still allowing for idiosyncratic stuff

Questions?

Towards an Equivalence Result

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- Question: is there a connection between potential factors driving returns and the SDF?
- Answer: Yes! Beta-representations derived from a factor model imply the existence of an affine SDF!

Theorem

Given the model

$$m = a + b'F, \quad \mathbb{E}[mR_i] = 1, \forall i \quad (1)$$

one can find $\gamma \in \mathbb{R}$ and $\lambda \in \mathbb{R}^K$ such that

$$\mathbb{E}[R_i] = \gamma + \beta_i' \lambda$$

where β_i are the regression coefficients of R_i on F (and a constant). Conversely, given γ and λ satisfying the beta-representation, one can find $a \in \mathbb{R}$ and $b \in \mathbb{R}^K$ such that the SDF in (1) prices all assets.

Proof - Part 1

We can assume without any loss of generality that $\mathbb{E}[F] = 0$ (otherwise redefine a accordingly).

Let's use the pricing equation in the following form:

$$\mathbb{E}[R_i] = \frac{1}{\mathbb{E}[m]} - \frac{\text{Cov}(m, R_i)}{\mathbb{E}[m]}$$

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Notice that $\mathbb{E}[m] = a$. Also,

$$\text{Cov}(m, R_i) = \text{Cov}(a + b'F, R_i) = b' \text{Cov}(F, R_i) = b' \text{Cov}[F, F] \text{Cov}[F, F]^{-1} \text{Cov}(F, R_i)$$

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If we define $\beta_i \equiv \text{Cov}[F, F]^{-1} \text{Cov}(F, R_i)$, we have

$$\mathbb{E}[R_i] = \frac{1}{a} - \frac{b' \text{Cov}[F, F] \beta_i}{a} = \gamma + \beta_i' \lambda$$

with $\gamma \equiv \frac{1}{a}$ and $\lambda \equiv -\frac{\text{Cov}[F, F]b}{a}$.

Proof - Part 2

Conversely, assume a beta-representation. Then, we can just define

$$a \equiv \frac{1}{\gamma}, \quad b \equiv -\text{Cov}[F, F]^{-1} \frac{\lambda}{\gamma}$$

which concludes the proof.

Important:

- We said nothing about uniqueness. This is not the unique SDF if markets aren't complete.
- The beta-representation is not necessarily unique either.

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Even more important:

$$\lambda = -\gamma \cdot \mathbb{E}[mF]$$

What's the intuition? λ is usually called the “factor risk premium”.

- (C): Chapters 6, 7, and 9

Hansen, Lars Peter, and Ravi Jagannathan. 1991. "Implications of Security Market Data for Models of Dynamic Economies." *Journal of Political Economy* 99 (2): 225–62.

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